7(a)(i)	A = 94, Z = 35	**[1]**
(ii)	Mass defect of U-235, m = (143)(1.00866) + (92)(1.00728) – (235.044) = 1.864 u Binding energy of U-235 = $mc^2 = (1.864)(1.66 \times 10^{-27})(3.00 \times 10^8)^2$ $= 2.785 \times 10^{-10}$ J = **1741 MeV** OR 1.864 (934) = 1741 MeV	**[1] sub for *m*** **[1] working** **[1] ans**
MC	Ecf within part	
(iii)	**Neutron(s) is/are produced**	**[1]**
(b)(i)	High energy **electron**	**[1]**
(ii)	$\lambda = \dfrac{\ln 2}{28(365)(24)(3600)} = 7.85 \times 10^{-10}\ \text{s}^{-1}$ $A = \lambda N = \lambda N_o e^{-\lambda t} = (7.85 \times 10^{-10})(2.36 \times 10^{13})\ e^{-\frac{\ln 2}{28}(100)}$ $= \mathbf{1.56 \times 10^3\ s^{-1}}$	**[1] sub for λ in s^{-1}** **[1] ans**
8(a)	The generator produces **alternating current**. The information provided in Fig. 8.3	**[1]**